

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 1: Computer Networks

Course: Computer Networks (Credit Hours: 3)

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1. WHAT IS A COMPUTER NETWORK?

A **Computer Network** means connecting two or more devices (computers, laptops, phones, printers) together using cables or Wi-Fi so they can talk, share files, and share hardware.

- **Sharing Resources:** Instead of buying 30 printers for 30 lab PCs, all PCs connect to **1 shared printer**.
- **Fast Communication:** Sending a file across the world in 1 second using the Internet or email.
- **Central Data Storage:** Keeping all company records safely on 1 central server instead of scattered across individual computers.

2. FIVE BASIC PARTS OF DATA COMMUNICATION

Whenever two devices communicate, they always need these 5 basic things:

1. Message (The Content): The actual information being sent (text message, WhatsApp voice note, photo, or PDF file).

2. Sender (Source): The device that creates and sends the message.

Easy Example Your smartphone when you press "Send" on WhatsApp.

3. Receiver (Destination): The device that gets and opens the message.

Easy Example Your friend's phone or computer screen receiving the message.

4. Medium (The Path / Road): The physical path through which data travels from sender to receiver.

- **Wired (Guided):** Ethernet LAN cable, Fiber optic cable.
- **Wireless (Unguided):** Wi-Fi signals, Mobile 4G/5G radio waves.

5. Protocol (The Rules & Language): A set of rules that both devices agree to follow.

Easy Example If a teacher speaks Chinese and students only know English, no communication happens even if sound travels fine. In networks, devices must speak the exact same language (like HTTP or TCP/IP) to understand each other.

3. ANALOG SIGNALS VS. DIGITAL SIGNALS

Feature	Analog Signals	Digital Signals
Shape	Smooth, continuous waves (sine wave ~).	Square, step-like pulses (0s and 1s 1).
Values	Infinite values (changes continuously).	Only 2 values: 0 (OFF / Low) or 1 (ON / High) .
Noise / Disturbance	Distorts easily (creates noise/hiss in sound).	Very clean; easy to fix distortion without loss.
Daily Example	Human voice speaking in a room; traditional FM radio.	Data stored inside your computer, laptop, or USB drive.

4. DIRECTION OF DATA FLOW (TRANSMISSION MODES)

1. Simplex (1-Way)

Traffic goes only in **one direction**.
The receiver can never send data back.

2. Half-Duplex (2-Way, 1 at a time)

Both can send and receive, but only **one person talks at a time**.

3. Full-Duplex (Both Together)

Both devices send and receive data **at the exact same time**.

Real-World Example Keyboard to CPU, or traditional TV broadcasting.

Real-World Example Walkie-Talkies (you speak, say "Over", then other speaks).

Real-World Example Mobile phone call (both can talk and hear simultaneously).

5. QUICK CLASS DISCUSSION QUESTIONS

Question 1: Why do computers use 0s and 1s (Digital) instead of human voice waves (Analog)?

Answer: Because 0s and 1s are immune to noise and easy for electronic chips to store accurately.

Question 2: When driving on a one-way bridge where cars can go both sides but only one car passes at a time, what mode is it?

Answer: **Half-Duplex.**